

AYMANE AIT DADS

Computer Vision & ML Intern | PyTorch · CLIP ViT-L/14 · Deep Learning

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Hiring Team - Summer 2026 Computer Vision & ML Intern- Master's/PhD

Applied Materials · Santa Clara, CA, United States

Dear Applied Materials Hiring Team,

Your team is developing **vision and ML-based software for semiconductor inspection and metrology** - work that demands models robust enough to detect defects at scales where human inspection fails. That precision requirement maps directly to what I built at EURECOM: I ranked **#1 on the private leaderboard** by fine-tuning **CLIP ViT-L/14 (428M parameters)** on 250K images spanning 25+ generator types, designing the system specifically to generalize to out-of-distribution inputs - the same challenge that makes wafer defect detection hard.

On the **deep learning algorithms** side, I ran systematic ablation studies across transformer block unfreezing depth (4, 6, and 8 layers), learning rate schedules, and augmentation strategies, proving that 1-epoch fine-tuning preserves generalization better than extended re-training - rigorous experimental discipline your algorithmic team requires. On the **Python prototyping** side, I implemented **FP16 mixed precision with GradScaler** and a **10-view test-time augmentation** pipeline end-to-end, shipping a full inference system optimized for real GPU constraints, not just benchmark numbers.

Applied Materials sits at the foundation of semiconductor manufacturing - the inspection and metrology systems your team builds directly determine whether next-generation chips meet yield targets that make advanced AI hardware possible. I would welcome the chance to contribute to that work in Santa Clara this summer and am available to discuss the role at your convenience.

Sincerely,

AYMANE AIT DADS